

Food Delivery Management System – ER Diagram Design

1. High-Level Overview

The Food Delivery Management System (FDMS) manages online food ordering, restaurant operations, courier delivery, and payments. It allows customers to browse restaurants, place orders, track deliveries, and make payments. Restaurants manage their menus and receive orders, while couriers deliver food efficiently. The system supports administrators for monitoring and issue resolution.

2. Entities, Attributes, and Relationships

Entity	Attributes (with Data Types)	Relationships
Customer	customer_id (INT, PK), name (VARCHAR), email (VARCHAR, UNIQUE), phone (VARCHAR), address (VARCHAR)	1:1 with Order
Restaurant	restaurant_id (INT, PK), name (VARCHAR), address (VARCHAR), phone (VARCHAR), rating (DECIMAL(2,1))	1:M with Menu_Item, 1:M with Order
Menu_Item	item_id (INT, PK), restaurant_id (INT, FK), item_name (VARCHAR), price (DECIMAL(6,2)), category (VARCHAR)	1:M with Order_Detail
Order	order_id (INT, PK), customer_id (INT, FK), restaurant_id (INT, FK), courier_id (INT, FK), order_date (DATETIME)	1:M with Order_Detail, 1:M with Payment
Order_Detail	order_id (INT, FK), item_id (INT, FK), quantity (INT), subtotal (DECIMAL(8,2))	PK (order_id, item_id), links Order and Menu_Item
Courier	courier_id (INT, PK), name (VARCHAR), phone (VARCHAR), vehicle_type (ENUM), availability_status (BOOLEAN)	1:M with Order
Payment	payment_id (INT, PK), order_id (INT, FK), payment_date (DATETIME), amount (DECIMAL(8,2)), payment_method (VARCHAR)	1:M with Order

3. ER Diagram

The diagram below illustrates entities, their attributes, and relationships in the Food Delivery Management System.

